

Enterprise Business Requirement Specification (BRS)

Project: Hotel Profile ETL Pipeline (AI-Driven Data Engineering Accelerator Input Document)

1. Objective

Design a production-grade ETL pipeline to ingest hotel profile data daily, detect changes (insert/update/delete), apply data quality validations, and maintain historical tracking using SCD Type 2.

2. Source System Details

Source Table: hotel_profile_source

Column	Type	Description
hotel_id	STRING	Primary key
hotel_name	STRING	Hotel name
city	STRING	City
country	STRING	Country
address	STRING	Address
rating	FLOAT	Rating (0-5)
active_flag	BOOLEAN	Active indicator
last_updated_ts	TIMESTAMP	Last update timestamp

3. Target Table Design (SCD Type 2)

Column	Type	Description
hotel_id	STRING	Business key
hotel_name	STRING	Hotel name
city	STRING	City
country	STRING	Country
address	STRING	Address
rating	FLOAT	Rating
is_active	BOOLEAN	Active flag
effective_start_date	TIMESTAMP	Start
effective_end_date	TIMESTAMP	End
is_current	BOOLEAN	Current record flag

4. Source to Target Mapping

Source Column	Target Column	Transformation
hotel_id	hotel_id	Direct
hotel_name	hotel_name	Trim + Standardize
city	city	Uppercase
country	country	Uppercase
rating	rating	Null check + range validation
active_flag	is_active	Direct

5. ETL Logic

1. Extract daily incremental data 2. Detect: - New records - Updated records (hash comparison) - Deleted records (soft delete) 3. Apply SCD Type 2 logic 4. Load into target table

6. Data Quality Rules

- 1 hotel_id not null
- 2 hotel_name not null
- 3 rating between 0 and 5
- 4 no duplicate hotel_id
- 5 timestamp not future

7. SLA & Performance

- Data freshness: Daily - Pipeline SLA: 2 hours - Partition: by ingestion_date

8. Architecture

Bronze → Raw ingestion Silver → Cleaned Gold → Business SCD table

9. Sample Record

hotel_id	hotel_name	city	rating
H1001	Taj Palace	Delhi	4.5